

Poisson equation and Sobolev spaces

calculations for week 12

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UAF Math 617 Functional Analysis

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Outline

- 1 the Poisson equation: $-\nabla^2 u = f$
- 2 the Poisson equation in weak form: $\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx$
- 3 a quick sketch of the finite element method
- 4 defining the Sobolev spaces: $H^k(\Omega)$
- 5 more about $H^1(\Omega)$
- 6 lecture content in week 12

the Poisson equation

- let $\Omega \subset \mathbb{R}^d$ be open; denote points by $x = (x_1, \dots, x_d) \in \Omega$
 - assume $d \in \{1, 2, 3\}$... I will draw pictures for $d = 2$
- assume the boundary of Ω is not too crazy (more below)
- assume $f \in L^2(\Omega)$ is (square-)integrable

Poisson equation

find $u = u(x)$ so that

$$-\nabla^2 u = f \quad \text{on } \Omega$$

- next slide defines “ ∇^2 ” ...
- picture:

the Laplacian

- given f , finding u from the Poisson equation $-\nabla^2 u = f$ is a problem of inverting the operator $-\nabla^2$

Definition (Laplacian operator)

for $\Omega \subset \mathbb{R}^d$ open and $u : \Omega \rightarrow \mathbb{R}$, the *Laplacian* of u is

$$\nabla^2 u = \nabla \cdot (\nabla u) = \frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_d^2}$$

- in each dimension:

$$d = 1 : \quad \nabla^2 u = u''$$

$$d = 2 : \quad \nabla^2 u = u_{xx} + u_{yy}$$

$$d = 3 : \quad \nabla^2 u = u_{xx} + u_{yy} + u_{zz}$$

- $\nabla^2 = \nabla \cdot \nabla$ usually appears in models because a flux is proportional to a gradient, and then a divergence connects interior behavior to a boundary integral of the flux
- we will need the divergence theorem from the [weeks 10&11 slides](#):

$$\int_{\Omega} \nabla \cdot \mathbf{V} \, dm = \int_{\partial\Omega} \mathbf{V} \cdot \mathbf{n} \, ds,$$

- recall that this theorem applies to $\Omega \subset \mathbb{R}^d$ if $\partial\Omega$ is Lipschitz or C^1
- when using this, $\mathbf{V}$ will be the gradient of a scalar function
- recall the boundary integral is the total flux of $\mathbf{V}$ outward through the boundary

Laplace's equation, and harmonic functions

- in the absence of any boundary conditions there is an infinite-dimensional space of solutions to $-\nabla^2 u = f$ on Ω

Definition (Laplace's equation = potential equation)

$$\nabla^2 u = 0$$

a solution of this $f = 0$ equation is called *harmonic*

- the set of harmonic functions on Ω is an ∞ -dimensional vector space of smooth functions:

$$\{w : \nabla^2 w = 0\} \subset C^\infty(\Omega)$$

- complex analysis is, essentially, the study of this space

boundary-value problems for the Poisson equation

- however, we want to build models which make *one* prediction
- the Poisson equation can be used to model heat conduction, electrostatic potential, elastic membranes, equilibrium distributions for random walks, or many other physical phenomena
- we will focus on *boundary-value problems* that pick-out one solution u

Dirichlet problem for the Poisson equation

for $f \in L^2(\Omega)$ and $g_D \in L^2(\partial\Omega)$, find $u : \Omega \rightarrow \mathbb{R}$ so that

$$\begin{aligned} -\nabla^2 u &= f && \text{on } \Omega \\ u &= g_D && \text{on } \partial\Omega \end{aligned}$$

boundary-value problems for the Poisson equation

- instead of u itself, the gradient ∇u could be set to known values along the boundary
- in models, the gradient is often proportional to a physical flux

Neumann problem for the Poisson equation

for $f \in L^2(\Omega)$ and $g_N \in L^2(\partial\Omega)$, find $u : \Omega \rightarrow \mathbb{R}$ so that

$$\begin{aligned} -\nabla^2 u &= f && \text{on } \Omega \\ \nabla u \cdot \mathbf{n} &= g_N && \text{on } \partial\Omega \end{aligned}$$

where $\mathbf{n}$ is the outward unit normal vector field along $\partial\Omega$

- other notation for the (scalar) directional derivative:

$$\nabla u \cdot \mathbf{n} = \frac{\partial u}{\partial n} = u_n$$

- one could have Dirichlet conditions on part of the boundary and Neumann on another part, as a disjoint union:

$$\partial\Omega = \partial_D\Omega \cup \partial_N\Omega$$

application: temperature of a conductive solid

- Fourier's law for heat conduction in solids says that the heat flux is

$$\mathbf{q} = -k\nabla u,$$

where u is the temperature, $\mathbf{q}$ is the heat flux, and $k > 0$ is the conductivity

- consider a solid with a heat capacity $c > 0$ (heat energy gained by increase in temperature) and a mass density $\rho > 0$ (mass per unit volume)
- given a heat source f within the domain Ω , conservation of energy says

$$c\rho \frac{\partial u}{\partial t} = -\nabla \cdot \mathbf{q} + f \quad \Longleftrightarrow \quad \frac{d}{dt} \left(\int_{\Omega} c\rho u \, dx \right) = - \int_{\partial\Omega} \mathbf{q} \cdot \mathbf{n} \, ds + \int_{\Omega} f \, dx$$

- assume steady state, so $\partial/\partial t = d/dt = 0$
- if k is constant then $-\nabla \cdot \mathbf{q} = -\nabla \cdot (-k\nabla u) = k\nabla^2 u$
- we derive Poisson's equation, the (steady) *heat* or *diffusion equation*

$$0 = k\nabla^2 u + f \quad \text{within } \Omega$$

- boundary conditions are needed! (coming soon)

application: deflection of an elastic membrane

- suppose $z = u(x, y)$ is a vertical displacement of an elastic membrane over a domain $\Omega \subset \mathbb{R}^2$
- suppose that along $\partial\Omega$ we have put a rigid frame with height $z = g(x, y)$
- suppose that the slopes along the graph of g are small
- suppose that the membrane is loaded by a force f (per unit area)
- group the elastic properties of the membrane into a constant $\kappa > 0$
- then the vertical displacement satisfies Poisson's equation

$$-\kappa \nabla^2 u = f \quad \text{within } \Omega$$

subject to $u = g$ along $\partial\Omega$

- the standard derivation starts from a the total elastic energy of the membrane, in which case the Poisson equation is the condition that the derivative is zero at the minimizer

applications: Poisson equation boundary value problems

pictures for $d = 1$ steady heat in a rod, with insulated ends:

pictures for $d = 2$ membrane over a frame:

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deriving the weak form of the Poisson equation

- first I'll describe *how* to get the weak formulation, then I'll explain *why*
- let's start from the Poisson equation, $-\nabla^2 u = f$
- we seek the solution function $u : \Omega \rightarrow \mathbb{R}$
- remember that $\nabla^2 = \nabla \cdot \nabla$, so the problem is:

$$-\nabla \cdot (\nabla u) = f \quad \text{within } \Omega$$

- suppose $v : \Omega \rightarrow \mathbb{R}$ is another function, which we call a *test function*
- multiply the Poisson equation by v and integrate over Ω :

$$-\int_{\Omega} \nabla \cdot (\nabla u) v \, dx \stackrel{*}{=} \int_{\Omega} f v \, dx$$

- apply the integration by parts rule from the [weeks 10&11 slides](#):

$$\int_{\Omega} \phi \nabla \cdot \mathbf{V} \, dm = \int_{\partial\Omega} \phi \mathbf{V} \cdot \mathbf{n} \, ds - \int_{\Omega} \nabla \phi \cdot \mathbf{V} \, dm$$

- we'll apply this on the left side of $*$, with $\phi = v$ and $\mathbf{V} = \nabla u$

deriving the weak form of the Poisson equation

- we get:

$$-\int_{\partial\Omega} v (\nabla u) \cdot \mathbf{n} \, ds + \int_{\Omega} \nabla v \cdot \nabla u \, dx = \int_{\Omega} f v \, dx$$

- rearrange to get:

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx + \int_{\partial\Omega} v (\nabla u \cdot \mathbf{n}) \, ds$$

- the integral $\int_{\Omega} f v \, dx$ is *data*, because f is given
- existence and uniqueness of the solution u , essentially requires turning the **boundary integral** into data also
- there are two options for removing u from the boundary integral:
 - 1 for Dirichlet boundary conditions, only use test functions v which have zero value along $\partial\Omega$
 - 2 for Neumann boundary conditions, replace $\nabla u \cdot \mathbf{n}$ with the data g_N

ALMOST the weak form of the Poisson equation

weak formulation of the Poisson equation

Let $\Omega \subset \mathbb{R}^d$ be an open, bounded domain with Lipschitz boundary, and suppose $f \in L^2(\Omega)$ is given. Solve one of these problems:

- 1 Find a solution $u \in C_c^1(\Omega)$ so that

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad (\text{homogeneous Dirichlet})$$

for all test functions $v \in C_c^1(\Omega)$.

- 2 Find a solution $u \in C^1(\Omega)$ so that

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx + \int_{\partial\Omega} g_N v \, ds \quad (\text{general Neumann})$$

for all test functions $v \in C^1(\Omega)$.

- this is not the right way to do it!
- asking for $C^1(\Omega)$ is too strong!

the weak form of the Poisson equation

weak formulation of the Poisson equation

Let $\Omega \subset \mathbb{R}^d$ be an open, bounded domain with Lipschitz boundary, and suppose $f \in L^2(\Omega)$ is given. Solve one of these problems:

- ① Find a solution $u \in H_0^1(\Omega)$ so that

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad (\text{homogeneous Dirichlet})$$

for all test functions $v \in H_0^1(\Omega)$.

- ② Find a solution $u \in H^1(\Omega)$ so that

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx + \int_{\partial\Omega} g_N v \, ds \quad (\text{general Neumann})$$

for all test functions $v \in H^1(\Omega)$.

- I will define the *Sobolev space* $H^1(\Omega)$ in the next section
- for general Dirichlet conditions there is an additional complication

what is the “weak formulation”?

observation 1.

The strong form is pointwise: $(-\nabla^2 u)(x) = f(x)$ holds at every $x \in \Omega$. By contrast, **the weak form is about the average using a test function**. That is, the weak form holds on the support of v .

- here's a cartoon of where the equation is enforced:

observation 2.

The strong form refers to the second derivatives, the Laplacian, of the unknown solution u . By contrast, **only the first derivatives of u** , namely the gradient, **is needed in the weak form**.

general Dirichlet boundary conditions

- Suppose $\Omega \subset \mathbb{R}^d$ is an open, bounded domain with Lipschitz boundary.
- Suppose $f \in L^2(\Omega)$ and $g_D \in L^2(\partial\Omega)$ are given.
- Recall the strong form of the general Dirichlet problem:

$$-\nabla^2 u = f \quad \text{on } \Omega, \quad \text{and } u = g_D \quad \text{on } \partial\Omega$$

- To build the general Dirichlet weak form we substitute $u + u_D$ into the homogeneous weak form.

weak form general Dirichlet problem

Let $u_D \in H^1(\Omega) \cap C(\bar{\Omega})$ be any function, *defined on the whole domain* Ω , so that $u_D = g_D$ along $\partial\Omega$. Find a solution $u \in H_0^1(\Omega)$ so that

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx - \int_{\Omega} \nabla u_D \cdot \nabla v \, dx$$

for all test functions $v \in H_0^1(\Omega)$.

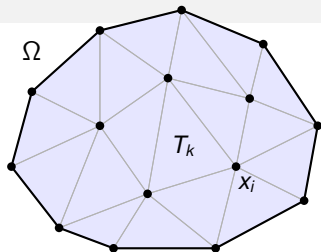
- The new right-hand side is data.

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what is a finite element method?

- suppose $\Omega \subset \mathbb{R}^2$ is a polygon
- put a mesh $\mathcal{T}$ of triangles on Ω
- denote nodes x_i and triangles T_k



- we call $f(x, y) = a + bx + cy$ a *linear polynomial* in two variables

Definition (the continuous P_1 finite element space over $\mathcal{T}$)

$$P_1 = \left\{ f \in C(\Omega) : f|_{T_k} \text{ is a linear polynomial} \right\}$$

- the approximate solution u_h , and the test functions v_h , will be from the following **finite-dimensional** subspace of V :

$$V_h = \{ f \in P_1 : f|_{\partial\Omega} = 0 \} \subset V$$

- $w_h \in V_h$ is determined by finitely-many numbers, the nodal values

what is a finite element method (FEM)?

weak form of the Poisson equation (homogeneous Dirichlet)

Find a solution $u \in V = H_0^1(\Omega)$ so that

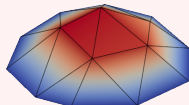
$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in V$$

- FEM solves the **same weak form**, but over the finite-dim'l subspace V_h

a FEM for the Poisson equation (homogeneous Dirichlet)

Find a solution $u_h \in V_h = \{f \in P_1 : f|_{\partial\Omega} = 0\}$ so that

$$\int_{\Omega} \nabla u_h \cdot \nabla v_h \, dx = \int_{\Omega} f v_h \, dx \quad \forall v_h \in V_h$$

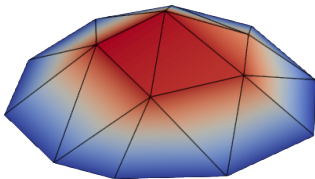


- a finite-dimensional vector $\mathbf{u} \in \mathbb{R}^N$ holds values of u_h at nodes x_i
- a computer program loops over the triangles T_k , computing the above integrals, to assemble a linear system in $\mathbb{R}^N$: $\mathbf{A}\mathbf{u} = \mathbf{b}$

finite element method automation

here is an FEM workflow:

- define a mesh in **Gmsh** format (`.msh`)
- solve the finite element problem using a Python program (`.py`) which calls the **Firedrake** library
- visualize the result using **Paraview** (`.pvd`)



blob.msh:

```
$MeshFormat
4.1 0 8
$EndMeshFormat
$PhysicalNames
2
1 1 "Boundary"
2 2 "Domain"
$EndPhysicalNames
$Entities
0 1 1 0
1 -0.5 0.0 0.0 4.0 3.5 0.0 1 1 0
1 -0.5 0.0 0.0 4.0 3.5 0.0 1 2 1 1
...
```

blob.py:

```
from firedrake import *
mesh = Mesh("blob.msh")
Vh = FunctionSpace(mesh, "P", 1)
uh = Function(Vh, name="u_h")
vh = TestFunction(Vh)
f = Constant(1.0)
F = dot(grad(uh), grad(vh)) * dx - f * vh * dx
zero = DirichletBC(Vh, Constant(0.0), (1,))
solve(F == 0, uh, bcs=[zero,])
VTKFile("result.pvd").write(uh)
```

see these codes at github.com/bueler/fa/tree/main/assets/slides/S26/codes

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recall weak derivatives

- given a bounded open set $\Omega \subset \mathbb{R}^d$
- recall the definition: $L^2(\Omega) = \{f : \Omega \rightarrow \mathbb{R} \mid \int_{\Omega} |f|^2 < \infty\}$
 - if Ω is bounded then $L^2(\Omega) \subset L^1(\Omega)$
- recall our multi-index derivative notation $D^{\alpha}\varphi$, from the [weeks 10&11 slides](#), and the main definition:

Definition (weak derivative)

if $f \in L^1_{\text{loc}}(\Omega)$, and if $g \in L^1_{\text{loc}}(\Omega)$ satisfies

$$\int_{\Omega} g \varphi \, dm = (-1)^{|\alpha|} \int_{\Omega} f D^{\alpha} \varphi \, dm \quad \text{for all } \varphi \in C_c^{\infty}(\Omega)$$

then we say g is the α th weak derivative of f , and we write $D^{\alpha}f = g$ a.e.

- example: f has a weak derivative $g = \partial f / \partial x_k \in L^1(\Omega)$ if

$$\int_{\Omega} g \varphi \, dm = - \int_{\Omega} f \frac{\partial \varphi}{\partial x_k} \, dm \quad \text{for all } \varphi \in C_c^{\infty}(\Omega)$$

$$\text{and } \int_{\Omega} |g| \, dm < \infty$$

Definition (Sobolev space $H^1(\Omega)$)

$$H^1(\Omega) = \left\{ f \in L^2(\Omega) \mid \begin{array}{l} \text{the weak partial derivatives } \frac{\partial f}{\partial x_k} \text{ exist} \\ \text{and are in } L^2(\Omega) \text{ for } k = 1, \dots, d \end{array} \right\}$$

is a Hilbert space with inner product:

$$\langle f, g \rangle_{H^1} = \int_{\Omega} f(x)g(x) \, dx + \int_{\Omega} \nabla f(x) \cdot \nabla g(x) \, dx$$

and norm: $\|f\|_{H^1} = \sqrt{\langle f, f \rangle_{H^1}} = \left(\int_{\Omega} |f|^2 + \int_{\Omega} |\nabla f|^2 \right)^{1/2}$

- *example on $\mathbb{R}^2$* : Consider the unit square $\Omega = (0, 1)^2$ and the function $f(x) = x_1 + x_2$. It has gradient $\nabla f = \langle 1, 1 \rangle$, and norm $\|f\|_{H^1} = \sqrt{19/6}$:

$$\begin{aligned} \|f\|_{H^1}^2 &= \int_{\Omega} |x_1 + x_2|^2 \, dx + \int_{\Omega} |\langle 1, 1 \rangle|^2 \, dx \\ &= \int_0^1 \int_0^1 x_1^2 + 2x_1x_2 + x_2^2 \, dx_1 \, dx_2 + \int_0^1 \int_0^1 2 \, dx_1 \, dx_2 = \frac{7}{6} + 2 = \frac{19}{6} \end{aligned}$$

Definition (Sobolev space $H^k(\Omega)$)

$$H^k(\Omega) = \left\{ f \in L^2(\Omega) \mid \begin{array}{l} \text{the weak derivatives } D^\alpha f \text{ exist} \\ \text{and are in } L^2(\Omega) \text{ for all } |\alpha| \leq k \end{array} \right\}$$

is a Hilbert space with inner product:

$$\langle f, g \rangle_{H^k} = \sum_{|\alpha| \leq k} \int_{\Omega} (D^\alpha f)(x) (D^\alpha g)(x) dx$$

and norm: $\|f\|_{H^k} = \sqrt{\langle f, f \rangle_{H^k}}$

- “ H ” is for Hilbert
- we may write $H^0(\Omega) = L^2(\Omega)$, because $D^\alpha f = f$ if $\alpha = (0, \dots, 0)$
- some sources use notation $H^k(\Omega) = W^{k,2}(\Omega)$
 - the “2” indicates that the weak derivatives are in $L^2(\Omega)$
 - general case: $W^{k,p}(\Omega)$ has weak derivatives in $L^p(\Omega)$ up to order k

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H^1 on an interval

- if $\Omega = (a, b)$ is an interval of the real line, then we have a simple definition using ordinary derivatives

Definition

$$H^1(a, b) = \left\{ f \in L^2(a, b) \mid \begin{array}{l} \text{the weak derivative } f' \text{ exists} \\ \text{and } f' \in L^2(a, b) \end{array} \right\}$$

- the norm is: $(\|f\|_{H^1})^2 = \int_a^b |f(x)|^2 dx + \int_a^b |f'(x)|^2 dx$
- but recall:
 - $L^2(a, b) \subset L^1(a, b)$, since $m((a, b)) < \infty$, and
 - $f' \in L^1(a, b) \implies f \in AC(a, b)$ and $f(x) = f(a) + \int_a^x f'(t) dt$
- most importantly:
$$H^1(a, b) \subset C^0([a, b])$$
 - and $C^1([a, b]) \subsetneq H^1(a, b) \subsetneq C^0([a, b]) \subsetneq L^2(a, b)$

H^1 on an interval, via Fourier series

recall Fourier series in $L^2(-\pi, \pi)$

using ON sequence $\frac{1}{\sqrt{2\pi}} e^{inx}$, $n \in \mathbb{Z}$, we have convergence in the $\|\cdot\|_2$ norm:

$$f(x) = \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} c_n e^{inx} \quad \text{with} \quad c_n = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} f(x) e^{-inx} dx,$$

Parseval's equality: $\|f\|_{L^2}^2 = \sum_{n=-\infty}^{\infty} |c_n|^2$

- but now consider the derivative formula: $f'(x) = \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} i n c_n e^{inx}$
- ... and we can norm the derivative (*use orthonormality*):

$$\int_{-\pi}^{\pi} |f'(x)|^2 dx = \sum_{k,n} \frac{1}{2\pi} \int_{-\pi}^{\pi} (i n c_n e^{inx}) (\overline{i k c_k e^{ikx}}) dx = \sum_{k,n} n k c_n c_k \delta_{kn} = \sum_{n=-\infty}^{\infty} n^2 |c_n|^2$$

Lemma ($H^1(-\pi, \pi)$ via Fourier series)

$$H^1(-\pi, \pi) = \left\{ f \in L^2(-\pi, \pi) \mid \sum_{n=-\infty}^{\infty} n^2 |c_n|^2 < \infty \right\}$$

H^k on an interval, via Fourier series

- the same logic applies for higher derivatives, e.g.:

$$\int_{-\pi}^{\pi} |f''|^2 = \sum_{k,n} \frac{1}{2\pi} \int_{-\pi}^{\pi} ((in)^2 c_n e^{inx}) (\overline{(ik)^2 c_k e^{ikx}}) dx = \sum_{n \in \mathbb{Z}} n^4 |c_n|^2$$

Lemma ($H^k(-\pi, \pi)$ via Fourier series)

$$H^k(-\pi, \pi) = \left\{ f \in L^2(-\pi, \pi) \mid \sum_{n \in \mathbb{Z}} n^{2k} |c_n|^2 < \infty \right\}$$

- with norm formula, by Parseval:

$$\begin{aligned} \|f\|_{H^k}^2 &= \int_{-\pi}^{\pi} |f|^2 + \cdots + \int_{-\pi}^{\pi} |f^{(k)}|^2 \\ &= \sum_{n \in \mathbb{Z}} (1 + n^2 + \cdots + n^{2k}) |c_n|^2 = \sum_{n \in \mathbb{Z}} \left(\sum_{j=0}^k n^{2j} \right) |c_n|^2 \end{aligned}$$

- we could make the lemma a definition, and **not need weak derivatives**
- also: **k does not need to be an integer**, but we will not pursue this

are functions in $H^1(\Omega)$ continuous for $\Omega \subset \mathbb{R}^2$?

- we have $H^1(a, b) \subset C^0([a, b])$, but this inclusion is dimension-dependent
- in examples below,

$$D = \{x \in \mathbb{R}^2 : |x|^2 < 1\}$$

is the unit disc, and $r = |x|$

- the first two examples are expected

a continuous function in $H^1(D)$... *expected*

$f(x) = \sqrt{r}$: $f \in C^0(D) \setminus C^1(D)$, and with polar coordinates $dm = r dr d\theta$,

$$\begin{aligned}\|f\|_{H^1}^2 &= \int_D |f|^2 + |\nabla f|^2 = 2\pi \int_0^1 \left[(\sqrt{r})^2 + \left(\frac{-1}{2\sqrt{r}} \right)^2 \right] r dr \\ &= 2\pi \int_0^1 \left(r^2 + \frac{1}{4} \right) dr = \frac{7\pi}{6} < \infty\end{aligned}$$

are functions in $H^1(\Omega)$ continuous for $\Omega \subset \mathbb{R}^2$?

a discontinuous function not in $H^1(D)$... *expected*

$g(x) = \ln(r)$: $g \notin C^0(D)$, and

$$\|g\|_{H^1}^2 = 2\pi \int_0^1 \left[(\ln(r))^2 + \left(\frac{1}{r}\right)^2 \right] r \, dr = (\text{finite}) + 2\pi \int_0^1 \frac{1}{r} \, dr = +\infty$$

are functions in $H^1(\Omega)$ continuous for $\Omega \subset \mathbb{R}^2$? NO

- define

$$p(x) = \begin{cases} \ln(|\ln(r)|), & 0 < r < 1/2 \\ \ln(\ln(2)), & 1/2 \leq r < 1 \end{cases}$$

a discontinuous function **which is in $H^1(D)$** ... *unexpected!*

$p(x)$: $p \notin C^0(D)$, and

$$\begin{aligned} \|p\|_{H^1}^2 &= (\text{finite}) + 2\pi \int_0^{1/2} \left(\frac{-1}{r|\ln(r)|} \right)^2 r \, dr = (\text{finite}) + 2\pi \int_0^{1/2} \frac{1}{r(\ln(r))^2} \, dr \\ &\stackrel{u=\ln(r)}{=} (\text{finite}) + 2\pi \int_{-\infty}^{-\ln(2)} \frac{du}{u^2} = (\text{finite}) + (\text{finite}) < \infty \end{aligned}$$

- $p \in H^1(D)$
 - computing exact values of L^2 and H^1 norms for g and p is on Assignment 7
- generally, $H^1(\Omega) \not\subset C^0(\Omega)$ **when $\Omega \subset \mathbb{R}^2$**
 - but $H^2(\Omega) \subset C^0(\Omega)$ does hold
- generally *Sobolev's* (and related) *inequalities* show which $H^k(\Omega)$, $\Omega \subset \mathbb{R}^d$, will contain continuous representatives

$H_0^1(\Omega)$ is the closed subspace with zero boundary values

- for any open domain $\Omega \subset \mathbb{R}^d$, the inclusion $C_c^1(\Omega) \subset H^1(\Omega)$ is clear

Definition

$$H_0^1(\Omega) = \overline{C_c^1(\Omega)}, \quad \text{where closure is in } H^1 \text{ norm}$$

- the 1D case already shows that $H_0^1(\Omega)$ is a proper closed subspace:
 $f(x) = 1 \in H^1(a, b)$ but $f \notin H_0^1(a, b)$ because of the FTC
 - $H_0^1(\Omega)$ is a proper closed subspace in any dimension
- picture:

fact: $\|\nabla f\|_{L^2}$ is equivalent to $\|f\|_{H^1}$ in $H_0^1(\Omega)$

- once you have zero boundary values, then the only way for the function to get large is for the gradient to also be large

Theorem (Poincaré-Friedrichs inequality)

Let $\Omega \subset \mathbb{R}^d$ be an open, bounded domain with Lipschitz boundary, or an open set which is bounded in one direction. There is $C > 0$, depending on Ω , so that

$$\|f\|_{L^2} \leq C \|\nabla f\|_{L^2} \quad \text{for all } f \in H_0^1(\Omega).$$

- main consequence: the simpler formula

$$|f|_{H^1} = \|\nabla f\|_{L^2} = \left(\int_{\Omega} |\nabla f|^2 \right)^{1/2}$$

is equivalent to the full norm $\|f\|_{H^1} = \left(\int_{\Omega} |f|^2 + \int_{\Omega} |\nabla f|^2 \right)^{1/2}$ on $H_0^1(\Omega)$

- it is *not* equivalent on the whole space $H^1(\Omega)$
- $|f|_{H^1}$ is called the H^1 seminorm
 - with the understanding that it is actually a norm on $H_0^1(\Omega)$

Poisson equation Dirichlet problems: weak forms

- now that $H_0^1(\Omega)$ is defined, we can make sense of these problems

weak formulation of Dirichlet problems

Let $\Omega \subset \mathbb{R}^d$ be an open, bounded domain with Lipschitz boundary, and suppose $f \in L^2(\Omega)$ is given. Solve:

- 1 **homogeneous Dirichlet:** Find $u \in H_0^1(\Omega)$ so that

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx$$

for all test functions $v \in H_0^1(\Omega)$.

- 2 **general Dirichlet:** Let $u_D \in H^1(\Omega) \cap C(\bar{\Omega})$ be any function so that $u_D = g_D$ along $\partial\Omega$. Find $\tilde{u} \in H_0^1(\Omega)$ so that

$$\int_{\Omega} \nabla \tilde{u} \cdot \nabla v \, dx = \int_{\Omega} f v \, dx - \int_{\Omega} \nabla u_D \cdot \nabla v \, dx$$

for all test functions $v \in H_0^1(\Omega)$. The solution is $u = \tilde{u} + u_D \in H^1(\Omega)$.

Outline

1. the Poisson equation: $-\nabla^2 u = f$
2. the Poisson equation in weak form: $\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx$
3. a quick sketch of the finite element method
4. defining the Sobolev spaces: $H^k(\Omega)$
5. more about $H^1(\Omega)$
6. lecture content in week 12

lecture content in week 12

- most of this material is **only from these slides**

definitions to know:

- Laplacian
- strong form of Poisson problems (Dirichlet and Neumann)
- weak form of Poisson problems (Dirichlet and Neumann)
- continuous P_1 finite element space over a mesh of triangles
- Sobolev spaces $H^k(\Omega)$
- examples showing $H^1(\Omega) \not\subset C^0(\Omega)$ for $\Omega \subset \mathbb{R}^2$
- $H_0^1(\Omega)$

propositions to know:

- lemma: characterize $H^k(-\pi, \pi)$ using Fourier series coefficients
- theorem: Poincaré-Friedrichs inequality for $H_0^1(\Omega)$
- Assignment 7 is posted at bueler.github.io/fa